

BG77xA-GL&BG95xA-GL& BG950S-GL PSM Application Note

LPWA Module Series

Version: 1.1

Date: 2024-08-29

Status: Released



At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2024. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2021-01-29	Igor STANCIC	Creation of the document
1.0	2022-01-14	Igor STANCIC/ Darko SAVIC	First official release
1.1	2024-08-29	Adam NAAS/ Nikola PERIC	1. Added applicable modules BG773A-GL, BG953A-GL, BG955A-GL and BG950S-GL (Chapter 1). 2. Added AT+CPSMS=2, which disables the use of PSM and restores default PSM configuration (Chapter 2.3.1). 3. Added a note clarifying the fact that PON_TRIG is disabled for modules with firmware version R02 (Chapter 3). 4. Added a note for AT commands not implemented for BG950S-GL (Chapter 3.2). 5. Added a note summarizing the different methods of waking up modules from PSM for modules with or without PON_TRIG function (Chapter 3.3).

Contents

About the Document.....	3
Contents	4
Table Index.....	5
Figure Index	6
1 Introduction	7
2 Description of PSM AT Commands	9
2.1. AT Command Introduction.....	9
2.1.1. Definitions.....	9
2.1.2. AT Command Syntax	9
2.2. Declaration of AT Command Examples.....	10
2.3. Description of AT Commands	10
2.3.1. AT+CPSMS Power Saving Mode Setting	10
2.3.2. AT+QPSMS Extended Power Saving Mode Setting	12
2.3.3. AT+QPSMCFG PSM Feature and Minimum Threshold Value Setting	14
2.4. URC Description	15
2.4.1. +QPSMTIMER Indicate the TAU Duration and Active Time Duration.....	15
3 Waking Up from PSM.....	16
3.1. Waking Up from PSM with PON_TRIG	16
3.2. Waking Up from PSM with PWRKEY	17
3.3. Waking Up from PSM When the T3412 Timer Expires	18
4 Typical Power Consumption Cycle.....	19
5 Appendix References	20

Table Index

Table 1: Applicable Modules..... 7

Table 2: Related Documents 20

Table 3: Terms and Abbreviations 20

Figure Index

Figure 1: Waking Up the Module from PSM with PON_TRIG	16
Figure 2: Waking Up the Module from PSM with PWRKEY	17

1 Introduction

This document describes the Power Saving Mode (PSM) feature and explains how to use the PSM function on the Quectel BG77xA-GL, BG95xA-GL and BG950S-GL modules.

The PSM feature enables IoT devices to stay inactive or powered down most of the time to save power. The PSM function wakes up the device only during data transmission, which usually happens for a short period of time. The PSM feature is especially important for eMTC/NB-IoT devices and applications that have the following characteristics:

- The eMTC/NB-IoT devices and applications are frequently inactive.
- Their active communication is of short duration.
- Data mainly originate from eMTC/NB-IoT devices (although the eMTC/NB-IoT devices can also download data when they are active).
- There are power constraints (devices running on a battery).
- The eMTC/NB-IoT devices require long battery life.

The PSM mode defined by 3GPP Release 13 is similar to power-off, but the UE (i.e., the module) remains registered on the network, so there is no need to re-attach or re-establish PDN connections. Therefore, the UE in PSM is not immediately available to mobile terminating services. The UE using PSM is available to mobile terminating services when it is in connected mode and during an Active Time that comes after the connected mode. The connected mode is caused by a mobile originated event like data transmission or signaling, e.g. after a periodic TAU procedure. The PSM is therefore suitable for UE that is expecting only infrequent mobile originating and terminating services, which can accept a corresponding latency in the mobile terminating communication.

Table 1: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL
-	BG950S-GL

NOTE

For more details regarding PSM mode defined by 3GPP Release 13, see **document [1]**.

2 Description of PSM AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of a corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. Description of AT Commands

2.3.1. AT+CPSMS Power Saving Mode Setting

The Write Command configures the PSM parameters of a module. It controls:

- whether the module wants to apply PSM or not,
- the requested extended periodic TAU value in E-UTRAN, and
- the requested Active Time value.

See the URCs provided by **AT+CEREG** for the Active Time value and the extended periodic TAU value that are allocated to the module by the network in E-UTRAN.

A special form of the command can be given as **AT+CPSMS=** (with all parameters omitted). In this form, the use of PSM is disabled (**<mode>=0**) and data for all parameters in the command are removed.

The Read Command returns the current parameter values.

The Test Command returns the supported **<mode>s**, the requested extended periodic TAU value in E-UTRAN and the requested Active Time value as compound values.

AT+CPSMS Power Saving Mode Setting	
Test Command AT+CPSMS=?	Response +CPSMS: (list of supported <mode>s),,(list of supported <Requested_Periodic-TAU>s),(list of supported <Requested_Active-Time>s) OK
Read Command AT+CPSMS?	Response +CPSMS: <mode> ,,,[<Requested_Periodic-TAU>],[<Requested_Active-Time>] OK

Write Command AT+CPSMS=[<mode>,,[,<Requested_Periodic-TAU>[,<Requested_Active-Time>]]]	Response OK If there is any other error: ERROR
Maximum Response Time	4000 ms
Characteristics	The command takes effect immediately. The configurations will be saved automatically.
Reference <i>3GPP TS 27.007</i>	

Parameter

<mode>	Integer type. Disable or enable the use of PSM in the UE. <u>0</u> Disable the use of PSM. 1 Enable the use of PSM. 2 Disable the use of PSM and restore default PSM configuration.
<Requested_Periodic-TAU>	String type. One byte in 8-bit format. Requested extended periodic TAU value (T3412) to be allocated to the UE in E-UTRAN (e.g. "00001010" equals 100 minutes). In the Write Command, when <mode> is 1 and <requested_periodicTAU> is omitted, the last configured value is used. Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 10 minutes. 0 0 1 value is incremented in multiples of 1 hour. 0 1 0 value is incremented in multiples of 10 hours. 0 1 1 value is incremented in multiples of 2 seconds. 1 0 0 value is incremented in multiples of 30 seconds. 1 0 1 value is incremented in multiples of 1 minute.
<Requested_Active-Time>	String type. One byte in 8-bit format. Requested Active Time value (T3324) to be allocated to UE (e.g., "00001111" equals 30 seconds). Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 2 seconds. 0 0 1 value is incremented in multiples of 1 minute. 0 1 0 value is incremented in multiples of decihours. 1 1 1 value indicates that the timer is deactivated, value cannot be set.

Example

```
AT+CPSMS=1,,"0000100","0001111" //Set the requested T3412 value to 40 minutes, and set the
OK                               requested T3324 value to 30 seconds.
```

WARNING

Every digital logic is powered down after the module enters PSM, so it is prohibited to provide any external voltage to the module's I/O ports that are not defined as a wake-up source. Pin VDD_EXT can be used to monitor PSM activity and control the external logic.

NOTE

If the debugging mode is enabled (**AT+QCFGEXT="debug"**), the module can refuse to enter PSM mode, depending on the logging activity load.

2.3.2. AT+QPSMS Extended Power Saving Mode Setting

This is an extended AT command, developed by Quectel, for setting PSM parameters. A special form of the command can be given as **AT+QPSMS=** (with all parameters omitted). In this form, the use of PSM is disabled (**<mode>=0**) and data for all parameters in the command are removed. It is similar to **AT+CPSMS**.

AT+QPSMS Extended Power Saving Mode Setting	
Test Command AT+QPSMS=?	Response +QPSMS: (list of supported <mode>s),,(list of supported <Requested_Periodic-TAU>s),(list of supported <Requested_Active-Time>s) OK
Read Command AT+QPSMS?	Response +QPSMS: <mode> ,,,[<Network_Periodic-TAU>],[<Network_Active-Time>] OK
Write Command AT+QPSMS=[<mode>,,[<Requested_Periodic-TAU>],[<Requested_Active-Time>]]]	Response OK If there is any other error: ERROR
Maximum Response Time	4000 ms
Characteristics	The command takes effect immediately. The configurations will be saved automatically.

Parameter

<mode>	Integer type. Disable or enable the use of PSM in UE. 0 Disable the use of PSM. 1 Enable the use of PSM.
<Requested_Periodic-TAU>	String type. One byte in 8-bit format. Requested extended periodic TAU value (T3412) to be allocated to UE in E-UTRAN (e.g., "00001010" equals 100 minutes). Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 10 minutes. 0 0 1 value is incremented in multiples of 1 hour. 0 1 0 value is incremented in multiples of 10 hours. 0 1 1 value is incremented in multiples of 2 seconds. 1 0 0 value is incremented in multiples of 30 seconds. 1 0 1 value is incremented in multiples of 1 minute.
<Requested_Active-Time>	String type. One byte in 8-bit format. Requested Active Time value (T3324) to be allocated to UE (e.g., "00001111" equals 30 seconds). Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 2 seconds. 0 0 1 value is incremented in multiples of 1 minute. 0 1 0 value is incremented in multiples of decihours. 1 1 1 value indicates that the timer is deactivated. Value cannot be set.
<Network_Periodic-TAU>	Integer type. Extended periodic TAU value (T3412) to be allocated to UE in E-UTRAN. The value is specified by the network.
<Network_Active-Time>	Integer type. Active timer value (T3324) to be allocated to UE in E-UTRAN. The value is specified by the network.

Example

```

AT+QPSMS=1,,"0000100","00001111" //Set the requested T3412 value to 40 minutes, and set the
                                     requested T3324 value to 30 seconds.

OK
AT+QPSMS?                             //Query the PSM mode, the periodic TAU and the active
                                     time specified by the network.

+QPSMS: 1,,"86400","2"

OK

```

WARNING

Every digital logic is powered down after the module enters PSM, so it is prohibited to provide any external voltage to the module's I/O ports that are not defined as a wake-up source. Pin VDD_EXT can be used to monitor PSM activity and control the external logic.

NOTE

If the debugging mode is enabled (**AT+QCFGEXT="debug"**), the module can refuse to enter PSM mode, depending on the logging activity load.

2.3.3. AT+QPSMCFG PSM Feature and Minimum Threshold Value Setting

The Write Command enables or disables the PSM feature and sets the minimum threshold value of the PSM cycle. **AT+QPSMCFG=** (with all parameters omitted) is a special form of the command that can be given. In this form, **<threshold>** is set to 20 and **<PSM_version>** is set to 4 (default).

AT+QPSMCFG PSM Feature and Minimum Threshold Value Setting	
Test Command AT+QPSMCFG=?	Response +QPSMCFG: (range of supported <threshold> s),(range of supported <PSM_version> s) OK
Read Command AT+QPSMCFG?	Response +QPSMCFG: <threshold> , <PSM_version> OK
Write Command AT+QPSMCFG=[<threshold>[,<PSM_version>]]	Response OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after rebooting. The configurations will be saved automatically.

Parameter

<threshold>	Integer type. Minimum threshold value of the PSM cycle. Condition to enter PSM: <threshold> less than PSM cycle (= T3412 - T3324). Range: <u>20</u> -4294967295. Unit: second.
--------------------------	---

<PSM_version>	Integer type. Bitmask to indicate the PSM feature (1: Enable; 0: Disable). Each bit is configured independently. Range: 0-4-15.
Bit 0	PSM without network coordination.
Bit 1	Rel-12 PSM without context retention.
Bit 2	Rel-12 PSM with context retention.
Bit 3	PSM in-between eDRX cycles.

Example

```

AT+QPSMCFG=100           //Set the threshold to 100 seconds.
OK
AT+QPSMCFG?               //Query the threshold value and the PSM version.
+QPSMCFG: 100,4
OK

```

2.4. URC Description

2.4.1. +QPSMTIMER Indicate the TAU Duration and Active Time Duration

The URC is used to indicate the TAU duration and active time duration of a module's PSM. The URC is disabled by default and can be enabled with **AT+QCFG="psm/urc",1**.

+QPSMTIMER Indicate the TAU Duration and Active Time Duration

+QPSMTIMER: <TAU_timer>,<T3324_timer>	Indicate the TAU duration and active time duration of a module's PSM.
--	---

Parameter

<TAU_timer>	Integer type. The TAU duration of PSM. Unit: second.
<T3324_timer>	Integer type. The active time duration of PSM. Unit: second.

Example

```

AT+QCFG="psm/urc"
+QCFG: "psm/urc",0
OK
AT+QCFG="psm/urc",1
OK

+QPSMTIMER: 86400,2           //The TAU and active timer values of PSM.

```

3 Waking Up from PSM

There are three scenarios to wake up the module:

- Forced: Drive PON_TRIG high to wake up the module.
- Drive PWRKEY low to wake up the module. The UART port stays inaccessible until PON_TRIG is driven high.
- Automatic: When the T3412 timer expires, the module wakes up automatically. The UART port stays inaccessible until the PON_TRIG is driven high.

NOTE

PON_TRIG is disabled on firmware version R02. R02 firmware refers to the firmware whose version number contains a segment “R02”, such as “BG950AGL...R02A01...”. The module firmware version can be obtained by the AT command **ATI** or **AT+GMR**. For more details about the commands, see *document [1]*.

3.1. Waking Up from PSM with PON_TRIG

To wake the module up from the PSM to realize communication between the module and the network.

Step 1: Drive PON_TRIG high and check if the module has been woken up from PSM.

Step 2: Establish the active UL and/or DL communication.

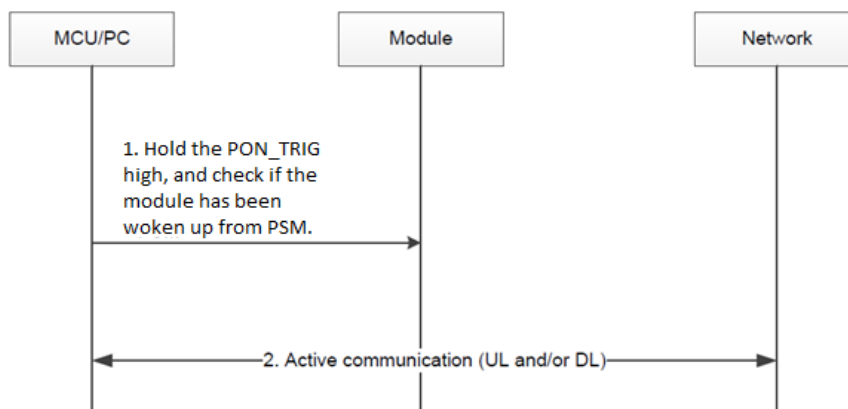


Figure 1: Waking Up the Module from PSM with PON_TRIG

3.2. Waking Up from PSM with PWRKEY

Like the PON_TRIG solution, the module can be woken up from PSM through steps below to realize communication between the module and the network.

Step 1: Drive PWRKEY low and check if the module has been woken up from PSM. The UART port stays inaccessible until the PON_TRIG is driven high.

Step 2: Establish the active UL and/or DL communication.

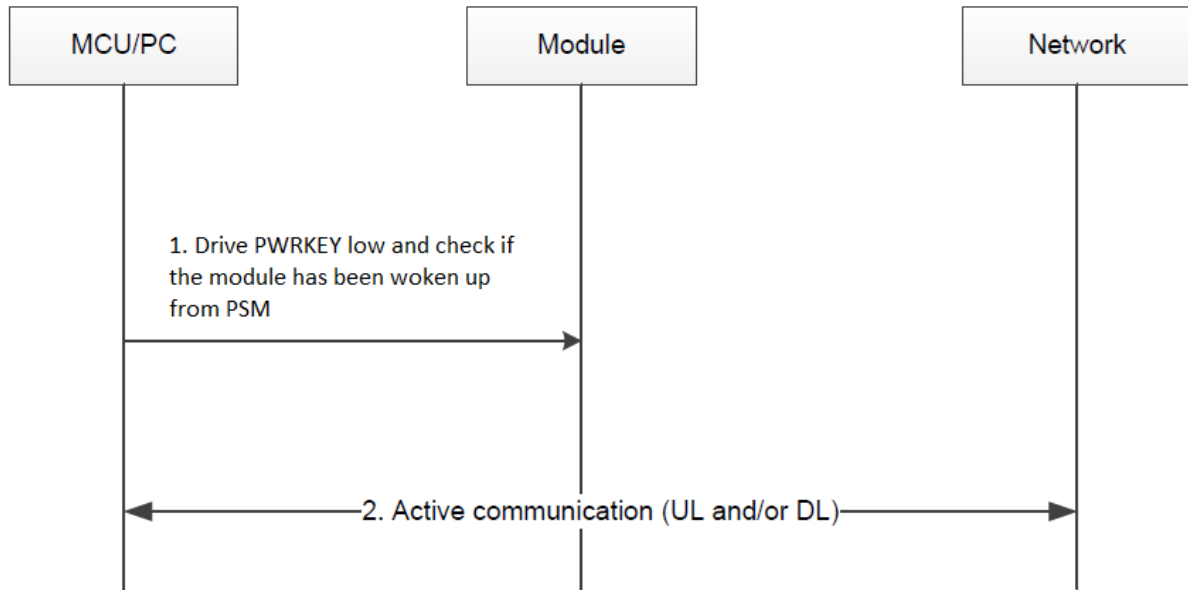


Figure 2: Waking Up the Module from PSM with PWRKEY

NOTE

This function is in development for BG950S-GL.

3.3. Waking Up from PSM When the T3412 Timer Expires

When the T3412 (extended TAU timer) expires, the module automatically wakes up from PSM.

The following figure illustrates the automatic procedure for waking up a module from PSM.

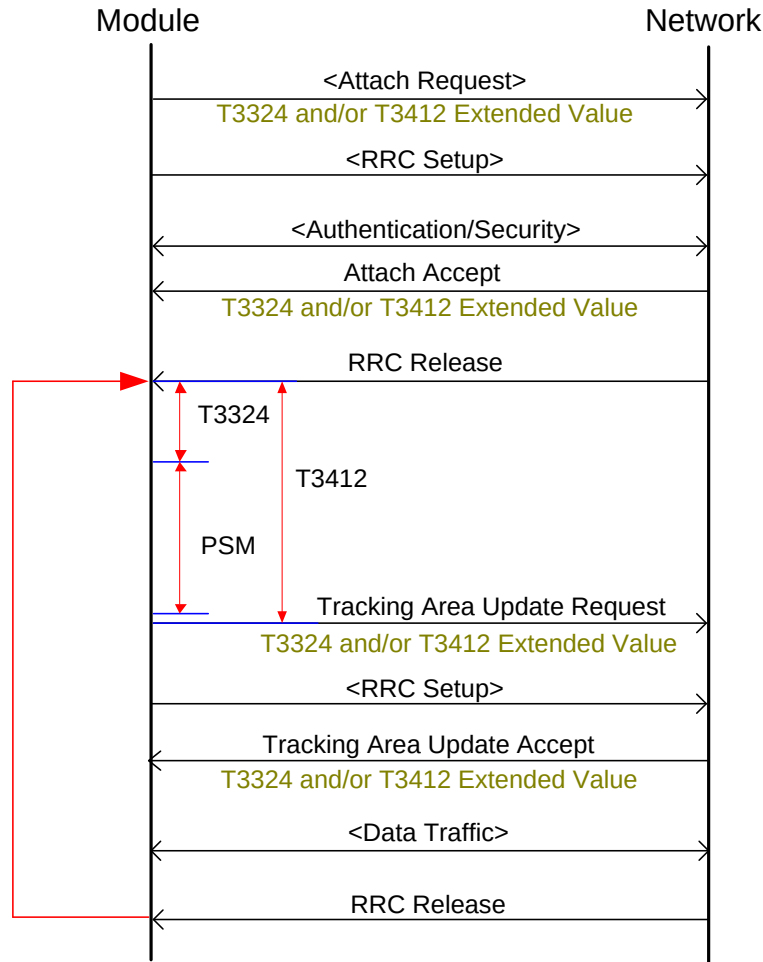


Figure 3: Waking Up the Module from PSM When T3412 Expire

NOTE

1. For modules with PON_TRIG function, any of the above methods can wake up the module from PSM.
2. For modules without PON_TRIG function, they can wake up in following cases:
 - When the TAU timer expires as described in **Chapter 3.3**.
 - When PWRKEY gets driven low as described in **Chapter 3.2**.

4 Typical Power Consumption Cycle

The following figure shows the typical power consumption cycle of the module.

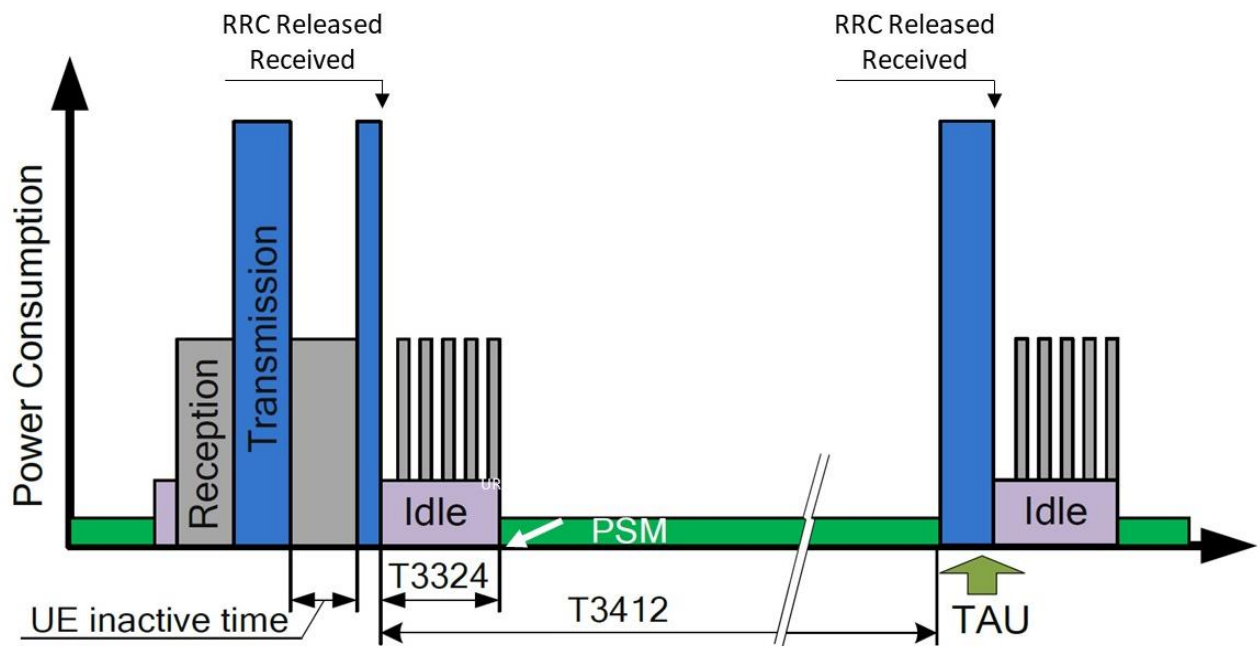


Figure 4: Typical Power Consumption Cycle

5 Appendix References

Table 2: Related Documents

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL&BG950S-GL_AT_Commands_Manual

Table 3: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
AP	Application Processor
DL	Downlink
eDRX	Extended Discontinuous Reception
eMTC	Enhanced Machine-Type Communication
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
MCU	Microcontroller Unit
NB-IoT	Narrow Band Internet of Things
OOS	Out Of Service
PC	Personal Computer
PDN	Packet Data Network
PSM	Power Saving Mode
RRC	Radio Resource Control
TAU	Tracking Area Update
UART	Universal Asynchronous Receiver/Transmitter
UE	User Equipment (typically, the module)
UL	Uplink
URC	Unsolicited Result Code